Fields and Waves

Exam 1

Fall 2025

1	**/10**
2	**/10**
3	**/10**
4	**/10**
Total	**/40**

Name _____

RIN _____

Notes:
SHOW ALL WORK. BEGIN WITH FORMULAS, THEN SUBSTITUTE VALUES AND UNITS. No credit will be given for numbers that appear without justification. Partial credit will be awarded where possible, so show your work so that we can follow your thought process. Use the backs of pages if there is not enough room on the front. If you do, please indicate that there is more work on the back of the page.

Unless otherwise stated in a problem, provide at least 2 significant digits in answers and include units with all numerical answers.

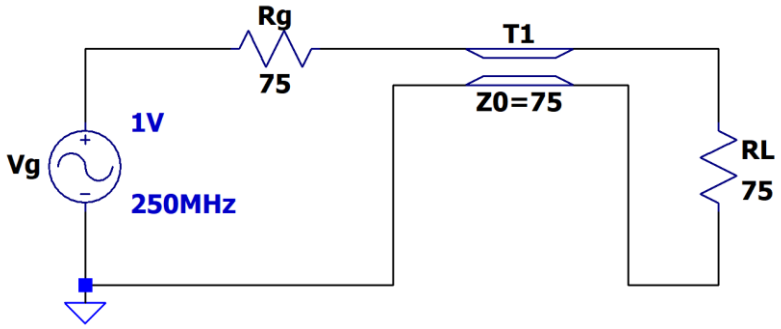
At least skim through the entire exam before you begin and then start with the problems you know best.

Problem 1: Cable TV Connection [10 pts]

You want to connect your television to a cable TV feed point located on the opposite side of your house. The only coaxial cable (T1) you have available for this purpose has the following properties:

- Length: 30 m
- Characteristic impedance: 75 Ω
- Velocity factor: 0.85

The cable TV feed point consists of a voltage source V_g with an amplitude of 1 V, source impedance R_g of 75 Ω and frequency f of 250 MHz. The television's load impedance R_L is also 75 Ω. A circuit diagram illustrating this setup is shown below.



a) [2 pts] If you use the provided cable to connect the cable TV feed point to your television, will there be a reflection at the load? Explain why or why not.

Since the load impedance is matched to the characteristic impedance, the reflection coefficient $\Gamma = 0$ and there will be no reflection.

b) [2 pts] What is the time delay of the coaxial cable?

$$T_d = \frac{L}{u} = \frac{30\ \text{m}}{0.85 * 3 \times 10^8\ \text{m/s}} = 1.176 \times 10^{-7}\text{s} = \mathbf{117.6\ ns}$$

c) [2 pts] What is the wavelength of the signal on the coaxial cable?

$$\lambda = \frac{u}{f} = \frac{0.85 * 3 \times 10^8\ \text{m/s}}{250 \times 10^6\ \text{Hz}} = \mathbf{1.02\ m}$$

d) [3 pts] Assuming that the coaxial cable can be considered low-loss and has a per-length series resistance of $r = 1\ \Omega/\text{m}$, calculate the amplitude of the *incident* voltage wave at the load.

$$\alpha = \frac{r}{2Z_0} = \frac{1\ \Omega/\text{m}}{2 * 75\Omega} = 6.667 \times 10^{-3}\ \text{m}^{-1}$$

$$V_0^+ = 1\text{V}\frac{75\Omega}{75\Omega + 75\Omega} = 0.5\text{V} = V(z = 0\text{m})$$

$$V(z = 30\text{m}) = V(z = 0\text{m})e^{-\alpha z} = 0.5\text{V} * e^{-6.667\times10^{-3}\text{m}^{-1}*30\text{m}} = \mathbf{0.409\ V}$$

e) [1 pt] If the coaxial cable's characteristic impedance were instead 50Ω, what is the minimum length of coaxial cable ($L > 0$) needed to ensure maximum power transfer from the source to the load?

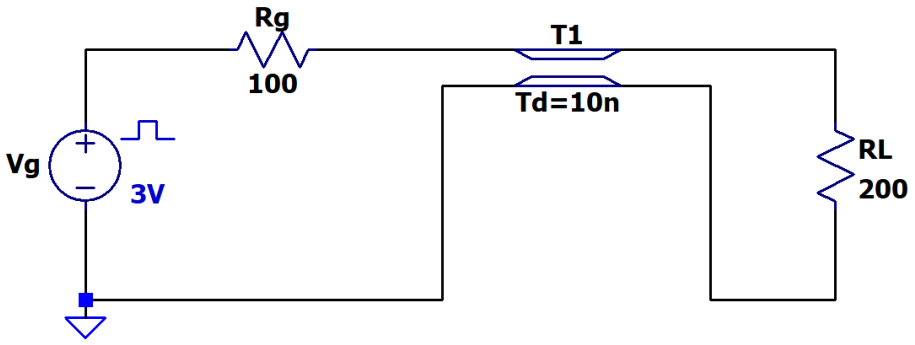
If $L = \lambda/2$, $Z_{in} = Z_L$, so $L = \frac{1.02\text{ m}}{2} = \mathbf{0.51\ m}$

Problem 2: Transmission Lines and Superconducting Qubits [10 pts]

In superconducting quantum computers, microwave voltage pulses are sent down transmission lines to perform operations on qubits. Consider one such transmission line with the following parameters:

- Per-length inductance: $l = 400$ nH/m
- Per-length capacitance: $c = 160$ pF/m
- Transit time (delay): $T_d = 10$ ns

At time $t = 0$, the voltage source sends a single DC pulse with amplitude 3 V and duration 1 ns down the transmission line toward the load. As a result of damage to the circuit, the values of the source impedance and load impedance have increased from their originally designed values to $R_g = 100\ \Omega$ and $R_L = 200\ \Omega$, as shown in the circuit schematic below.



a) [1 pt] Calculate the characteristic impedance of the transmission line.

$$Z_0 = \sqrt{l/c} = \sqrt{(400 \times 10^{-9}\ \text{H/m})/(160 \times 10^{-12}\ \text{F/m})} = \mathbf{50\ \Omega}$$

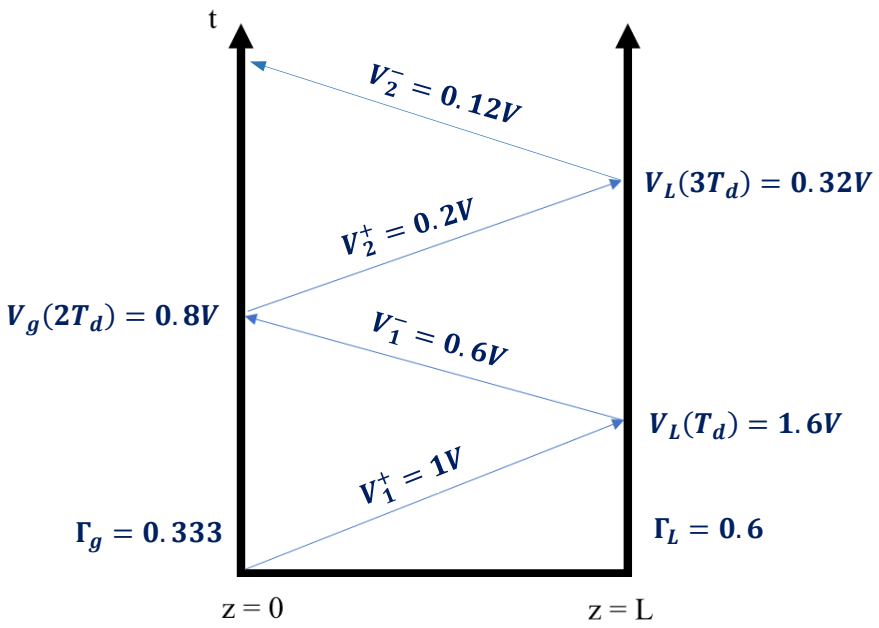
b) [2 pts] What is the initial amplitude of the voltage pulse V_1^+ that makes it onto the transmission line?

$$V_1^+ = V_g \frac{Z_0}{R_g + Z_0} = 3\text{V}\frac{50\ \Omega}{100\ \Omega + 50\ \Omega} = \mathbf{1\ V}$$

c) [6 pts] Sketch a bounce diagram for the voltage amplitude of the pulse for time $0 \le t < 4T$. Be sure to label both traveling voltage amplitudes and the amplitudes that would be measured at the load and the source during reflection. You do not need to calculate the total voltage at the source for $t = 4T_d$.

Reflection coefficient at the load: $\Gamma_L = \frac{200\ \Omega - 50\ \Omega}{200\ \Omega + 50\ \Omega} = 0.6$

Reflection coefficient at the generator: $\Gamma_g = \frac{100\ \Omega - 50\ \Omega}{100\ \Omega + 50\ \Omega} = 0.333$

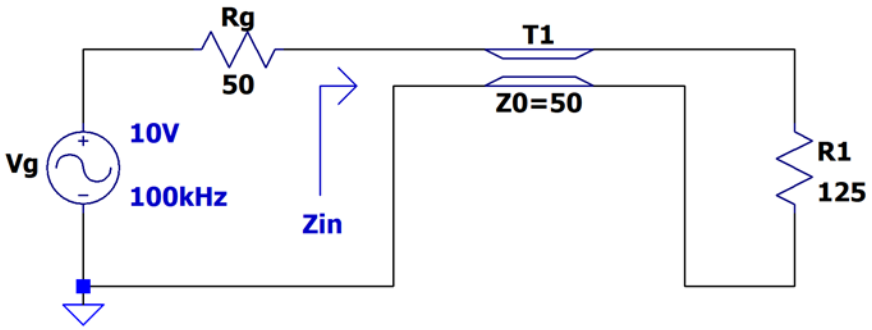


d) [1 pt] What will be the voltage on the transmission line after an infinite number of bounces (t→ ∞)?

The voltage on the transmission line will be **0 V**, since the pulse will transmit voltage to the source and load resistances on each reflection until the amplitude of the pulse becomes zero.

Problem 3: Quarter Wave Transmission Lines [10 pts]

The transmission line in the circuit below has a characteristic impedance $Z_0 = 50\Omega$ and length $L = \frac{\lambda}{4}$. The amplitude of the sinusoidal source voltage is $V_g = 10$ V, its wavelength is $\lambda = 20$ m and its frequency is $f = 100$ kHz.



a) [1 pt] Calculate the input impedance of the transmission line.

Since this is a quarter-wave transformer, $Z_{in} = \frac{Z_0^2}{Z_L} = \frac{(50\ \Omega)^2}{125\Omega} = \mathbf{20\ \Omega}$

b) [4 pts] Write down the phasor for the total voltage wave $\tilde{V}$(z) on the transmission line.

$$\tilde{V} = V_0^+ e^{-j\beta z} + V_0^- e^{j\beta z} = V_0^+ \left(e^{-j\beta z} + \Gamma_L e^{j\beta z}\right)$$

$$V_0^+ = 10V \frac{50\Omega}{50\Omega + 50\Omega} = 5\ \text{V}$$

$$\Gamma_L = \frac{125\ \Omega - 50\ \Omega}{125\ \Omega + 50\ \Omega} = 0.429$$

$$\beta = \frac{2\pi}{\lambda} = \frac{2\pi}{20\ \text{m}} = 0.314\ \text{m}$$

$$\tilde{V}(z) = 5\left(e^{-j0.314z} + 0.429e^{j0.314z}\right)\ \text{V} = \mathbf{5e^{-j0.314z} + 2.145e^{j0.314z}\ V}$$

c) [2 pts] Convert your expression for $\tilde{V}$(z) to the time domain $v(z,t)$.

$$v(z,t) = \text{Re}\left[\tilde{V}(z)e^{j\omega t}\right] = \text{Re}\left[\left(5e^{-j0.314z} + 2.145e^{j0.314z}\right)e^{j6.28\times10^5 t}\ \text{V}\right]$$

$$= \mathbf{5\cos\left(6.28\times10^5 t - 0.314z\right) + 2.145\cos\left(6.28\times10^5 t + 0.314z\right)\ V}$$

d) [2 pts] In order to deliver maximum power to the load with a transmission line of length $L = \lambda/4$, you would need a transmission line with a different characteristic impedance. What would the characteristic impedance of this new transmission line need to be?

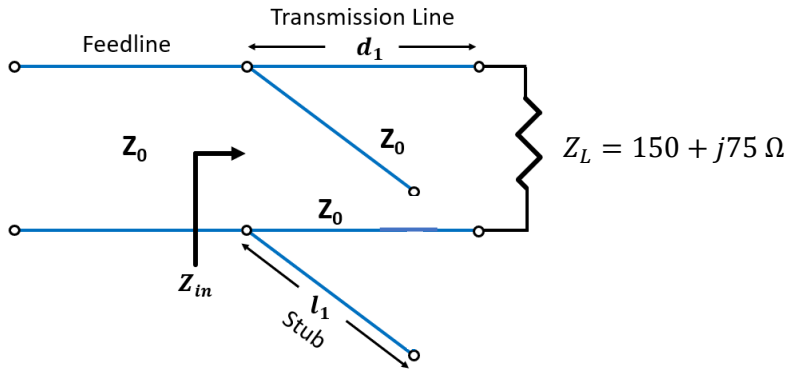
$$Z_{in} = \frac{Z_0^2}{Z_L} \rightarrow 50\ \Omega = \frac{Z_0^2}{125\ \Omega} \rightarrow Z_0 = \sqrt{50\Omega * 125\Omega} = \mathbf{79.06\ \Omega}$$

e) [1 pt] Even though the transmission line you chose in part (d) delivers maximum power to the load in steady state, reflections will still occur on the transmission line at the load. How can there be maximum power transfer, but also reflections on the transmission line?

In steady state, the total voltage and current waves on the transmission line result from the sum of the incident and reflected waves, which interfere with each other to create an impedance that varies along the transmission line. By choosing the transmission line length appropriately, the input impedance Z_{in} can be made to be $50\ \Omega$, enabling the source to deliver maximum power to the load, even though reflections still occur at the load.

Problem 4: Single-Stub Matching [10 pts]

You have been asked to match the load impedance $Z_L = 150 + j75\ \Omega$ to a feedline with characteristic impedance $Z_0 = 75\ \Omega$ by using an open-circuited transmission line stub. A diagram of the single-stub matching circuit is shown below. A Smith chart is attached to the back of your exam for recording your answers. A ruler will also be provided.



a) [2 pts] Calculate the normalized load impedance and plot it on the Smith chart.

$$z = \frac{Z_L}{Z_0} = \frac{150+j75\ \Omega}{75\ \Omega} = \mathbf{2 + j}\ \textbf{(see Smith chart)}$$

b) [2 pts] Plot the normalized load admittance on the Smith chart and show how you determined it from the plotted normalized load impedance. Write down the normalized load admittance you found from the Smith chart.

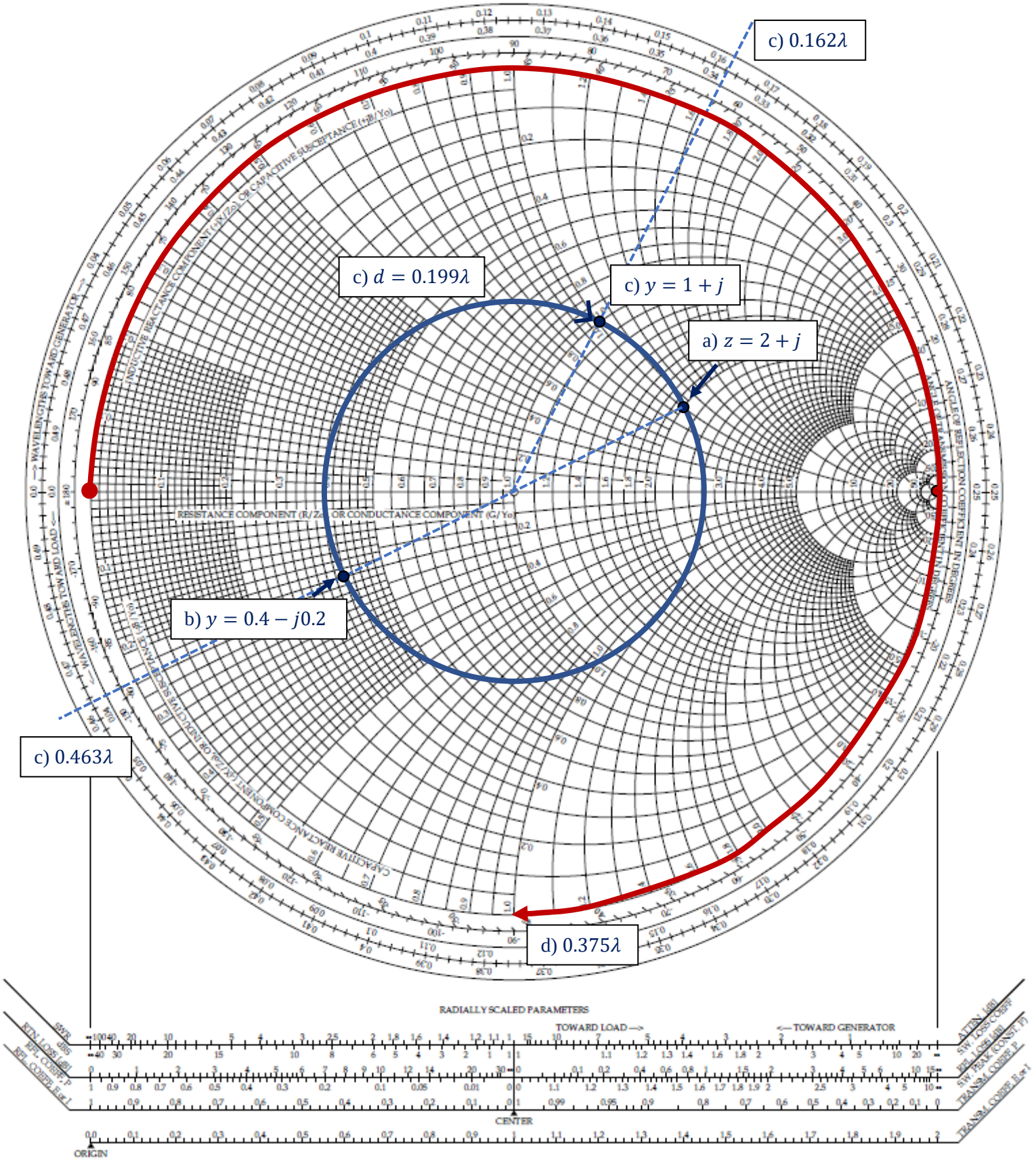
$$\boldsymbol{y = 0.4 - j0.2}\ \textbf{(see Smith chart)}$$

c) [3 pts] Calculate the distance d_1 (in terms of wavelength) from the load where the open-circuited stub should be placed, and give the input admittance y_d of the transmission line at that point. Show all necessary work on the Smith chart for how you determined this value.

$$d = (0.5\lambda - 0.463\lambda) + 0.162\lambda = \boldsymbol{0.199\lambda}\ \textbf{(see Smith chart)}$$

d) [3 pts] Calculate the length l_1 (in terms of wavelength) of the stub needed to match the source, and give the input admittance y_s of the open-circuited stub of that length. Show all necessary work on the Smith chart for how you determined this value.

$$l = \boldsymbol{0.375\lambda}\ \textbf{(see Smith chart)}$$


c) 0.162λ
c) d = 0.199λ
c) y = 1 + j
a) z = 2 + j
b) y = 0.4 − j0.2
c) 0.463λ
d) 0.375λ
RESISTANCE COMPONENT (R/Z₀) OR CONDUCTANCE COMPONENT (G/Y₀)
INDUCTIVE REACTANCE COMPONENT (+jX/Z₀) OR CAPACITIVE SUSCEPTANCE (+jB/Y₀)
CAPACITIVE REACTANCE COMPONENT (-jX/Z₀) OR INDUCTIVE SUSCEPTANCE (-jB/Y₀)
—> WAVELENGTHS TOWARD GENERATOR —>
<— WAVELENGTHS TOWARD LOAD <—
ANGLE OF REFLECTION COEFFICIENT IN DEGREES
RADIALLY SCALED PARAMETERS
TOWARD LOAD —>
<— TOWARD GENERATOR
CENTER
ORIGIN